Lecture 4: Basic Computability Theory, Recursive Functions

Models of Computation

https://clegra.github.io/moc/moc.html

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Course overview

intro	*classic models*			*additional models*
Introduction to Computability	**Machine Models**	**Recursive Functions**	**Lambda Calculus**	**Three more Models of Computation**
computation and decision problems, from logic to computability, overview of models of computation relevance of MoCs	Post Machines, typical features, Turing's analysis of human computers, Turing machines, basic recursion theory	primitive recursive functions, Gödel–Herbrand recursive functions, partial recursive funct's, partial recursive = = Turing-computable, Church's Thesis	λ-terms, β-reduction, λ-definable functions, partial recursive = λ-definable = Turing computable	Post's Correspondence Problem, Interaction-Nets, Fractran
	imperative programming	*algebraic programming*	*functional programming*	

Overview

- Elementary recursion theory
  - an unsolvable problem
  - Halting problem
  - universal language
  - Chomsky hierarchy

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- Elementary recursion theory
  - an unsolvable problem
  - Halting problem
  - universal language
  - Chomsky hierarchy
- Recursive functions
  - primitive recursive functions
  - Gödel–Herbrand(–Kleene) general recursive functions
  - partial recursive functions
    - defined with μ-recursion (unbounded minimisation)
  - Partial recursive functions = Turing computable functions
  - Church's thesis
    - effectively calculable functions $\hat{=}$ partial-recursive functions
    - some debate

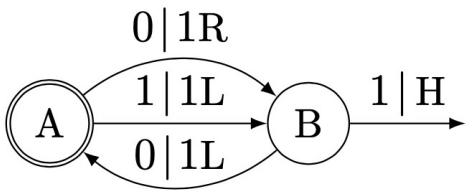
Busy Beavers

Definition (Tibor Radó, 1962)

$BB(n) :=$ the largest number of steps any n-state Turing machine with tape alphabet $\{0, 1\}$ and blank symbol 0 can run before eventually halting, when started on an empty tape.

$BB(1) = 1$	(easy)
$BB(2) = 6$	(homework exercise)

2-State Busy Beaver



initial state	0	0	0	0A	0	0	
step 1	0	0	0	1	0B	0	
step 2	0	0	0	1A	1	0	
step 3	0	0	0B	1	1	0	
step 4	0	0A	1	1	1	0	
step 5	0	1	1B	1	1	0	
step 6 (halt)	0	1	1H	1	1	0	

Figure 1: A 2-state Busy Beaver and its execution on an initially all-0 tape. Starting in state A, the machine finds a 0 on its tape and therefore follows the arrow labeled $0|1R$, which causes it to replace the 0 by a 1, move one square to the right, and transition into state B, and so on until the machine reaches the Halt arrow on its 6th step.

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BB$(2) = 6$	(homework exercise)
BB$(3) = 21$	(S. Lin and T. Radó, 1965)
BB$(4) = 107$	(A. Brady, 1983)
BB$(5) = \begin{cases} \geq 47176870 \\ = 47176870 \end{cases}$	(1990) (international team, 2024)

$$BB(6) \geq \underbrace{10^{10^{\cdot^{\cdot^{\cdot^{10}}}}}}_{15}$$

- Scott Aaronson: "The Busy Beaver Frontier", [1].
- —"Why Philosophers Should Care About Computational Complexity?" [2]

Elementary Recursion Theory

An unsolvable problem

The diagonalisation language:

$$L_d := \{w \mid w = \langle M \rangle, w \notin L(M)\} = \{\langle M \rangle \mid \langle M \rangle \notin L(M)\}$$

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L_d is not recursively enumerable.

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Membership in the diagonalisation language
Instance: w a binary word.
Question: Does $w \in L_d$ hold? (Does Tm. M with $\langle M \rangle = w$ accept w?)

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Theorem

There exist unsolvable decision problems.

Exercise: Halting Problem

Exercise

Try to adapt the diagonalisation argument to show that for the Halting Problem

$$H = \{ \langle \langle M \rangle , w \rangle \mid M \text{ halts on input } w \}$$

it holds:

- H is not recursive

and show that:

- H is recursively enumerable

Properties of r.e./recursive sets (I)

For $L \subseteq \Sigma^*$, $\bar{L} := \Sigma^* \setminus L$ is called the complement of L.

Proposition

If L is recursive, then $\bar{L}$ is recursive.

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First idea: Swap the accepting states of M with the non-accepting states of M in which computations may halt.

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M is modified as follows to obtain $\bar{M}$:

- the accepting states of M are made non-accepting in $\bar{M}$.
- $\bar{M}$ has a new accepting state r.
- for each $q \in Q$ and tape symbol $s \in \Gamma$ such that $\delta_M(q, s)$ is undefined, add the transition $\delta_{\bar{M}}(q, s) = \langle r, s, R \rangle$.

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It follows that $\bar{L} = L(\bar{M})$, and that $\bar{M}$ halts on all inputs. $\square$

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Proposition

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Proof.

Let M_1 and M_2 be Tm's such that $L = L(M_1)$ and $\bar{L} = L(M_2)$.

To decide, for a given $w \in \Sigma^*$, whether $w \in L$, build a Tm M that executes M_1 and M_2 on w in parallel, and such that:

- if M_1 accepts w, then also M accepts w.
- if M_2 accepts w, then also M halts, but does not accept w.

Hence M accepts w iff $w \in L(M_1) = L$. Thus $L(M) = L$.

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Hence M accepts w iff $w \in L(M_1) = L$. Thus $L(M) = L$.

Since for all w, either $w \in L$ or $w \in \bar{L}$, it follows that either M_1 or M_2 halts on w, and hence M halts on all inputs.

Hence $L = L(M)$ is recursive.

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Universal language

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- L_u is r.e.: $L_u = L(M_u)$ for an universal machine M_u.

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Proof.

- L_u is r.e.: $L_u = L(M_u)$ for an universal machine M_u.
- L_u is not recursive:

  Suppose that L_u is recursive. Then $\bar{L_u}$ is recursive, and hence there exists a Tm. M such that $\bar{L_u} = L(M)$.

  M can be used to build a Tm. M' that accepts the diagonalisation language L_d, entailing $L_u = L(M')$.

  [picture of M' to be given]

  But then L_u would actually be r.e., in contradiction with what we proved before.

Busy Beaver is not computable

Definition (Tibor Radó, 1962)

$BB(n) :=$ the largest number of steps any n-state Turing machine with tape alphabet $\{0, 1\}$ and blank symbol 0 can run before eventually halting, when started on an empty tape.

Proposition

The busy beaver function $BB : \mathbb{N} \to \mathbb{N},\ n \mapsto BB(n)$ *is not (Turing-)computable.*

More generally: one can solve the Halting Problem, given oracle access to any function $b : \mathbb{N} \to \mathbb{N}$ *such that* $b(n) \geq BB(n)$ *for all* $n \in \mathbb{N}$.

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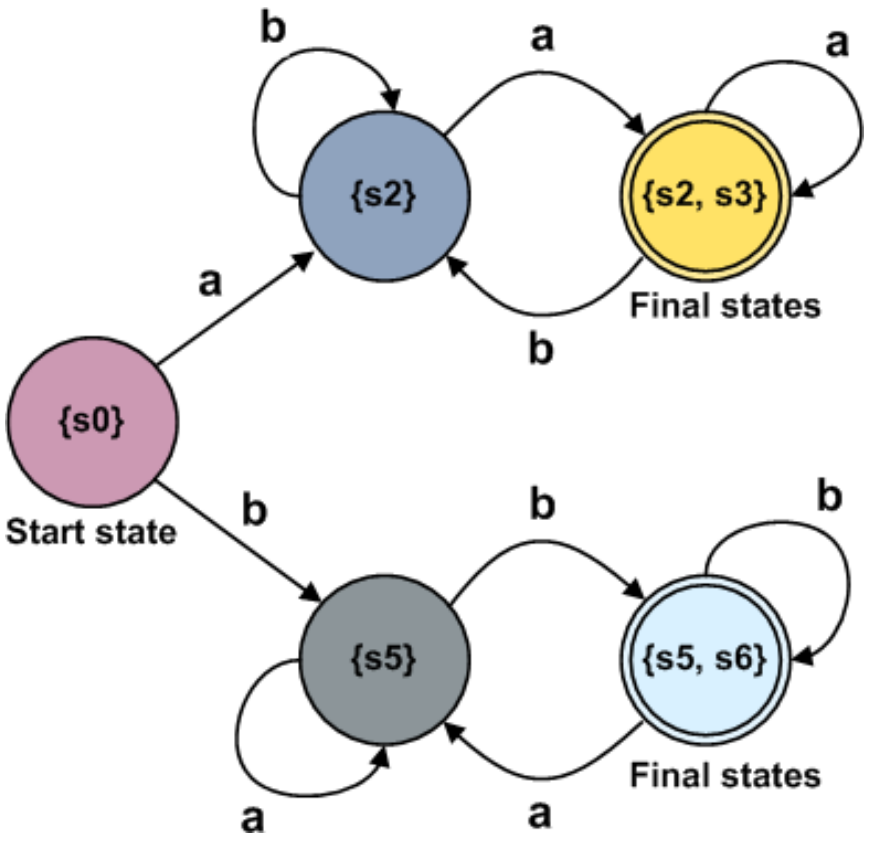
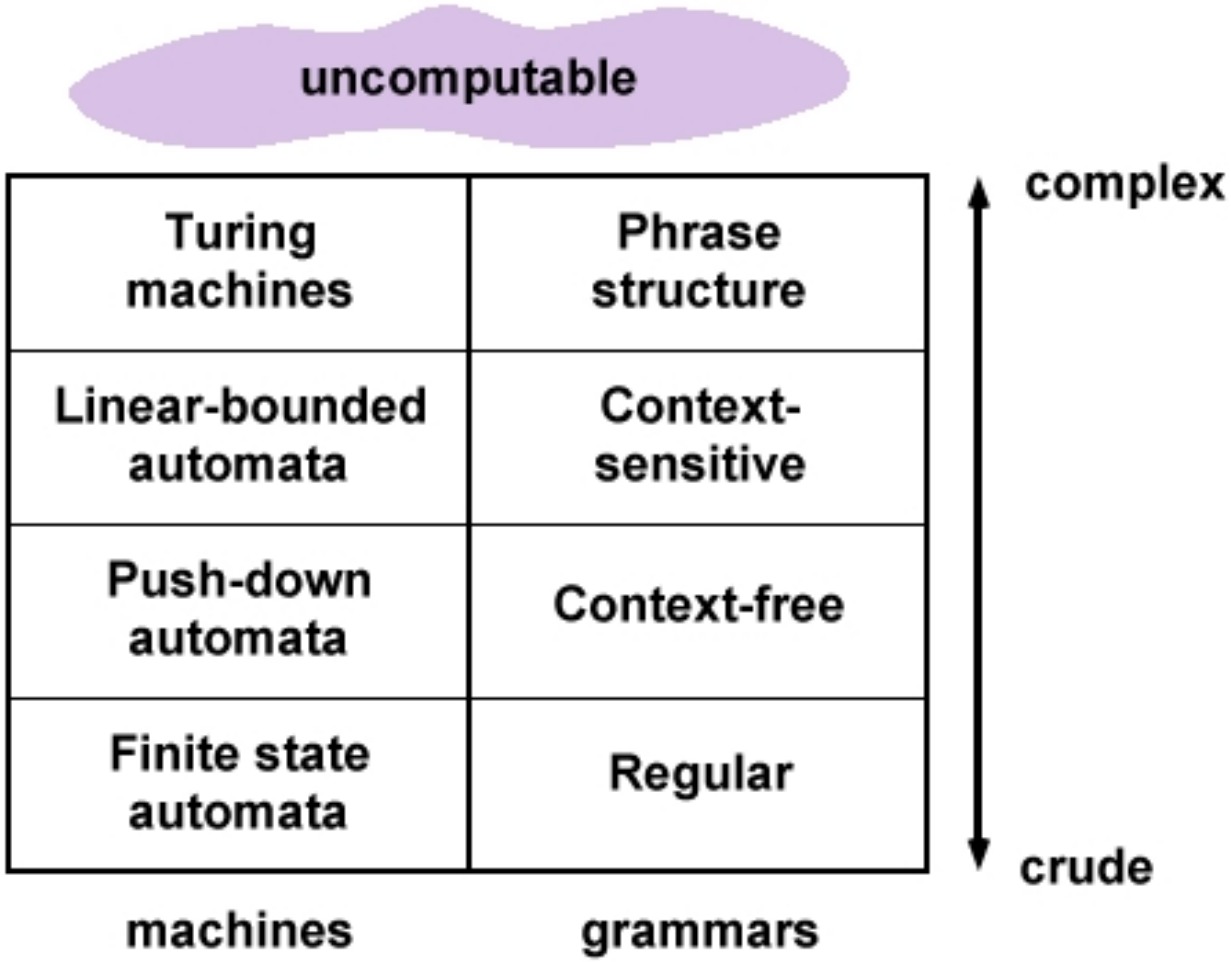
Proposition

Let $f : \mathbb{N} \to \mathbb{N}$ *be a Turing-computable function.*
Then there exists an $n_f \in \mathbb{N}$ *such that* $BB(n) > f(n)$ *for all* $n \geq n_f$.

Finite-state automaton



Formal-languages Chomsky hierarchy



Calculable function?

Questions/Exercises

2. Let $f : \mathbb{N} \to \mathbb{N}$ defined by

$$n \longmapsto \begin{cases} 0 & \ldots\ n = 0\ \&\ \text{Goldbach's conjecture is false} \\ 1 & \ldots\ n = 0\ \&\ \text{Goldbach's conjecture is true} \\ n+1 & \ldots\ n > 0 \end{cases}$$

Is f calculable?

Calculable function?

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Is f calculable?

Answer: Yes, because it is one of two calculable functions.
(We just do no know which one.)

Timeline: From logic to computability

1900	Hilbert's 23 Problems in mathematics
1910/12/13	Russell/Whitehead: Principia Mathematica
1928	Hilbert/Ackermann: formulate completeness/decision problems for the predicate calculus (the latter called 'Entscheidungsproblem')
1929	Presburger: completeness/decidability of theory of addition on $\mathbb{Z}$
1930	Gödel: completeness theorem of predicate calculus
1931	Gödel: incompleteness theorems for first-order arithmetic
1932	Church: λ-calculus
1933/34	Herbrand/Gödel: general recursive functions
1936	Church/Kleene: λ-definable $\sim$ general recursive Church Thesis: 'effectively calculable' be defined as either Church shows: the 'Entscheidungsproblem' is unsolvable Post: machine model; Church's thesis as 'working hypothesis'
1937	Turing: convincing analysis of a 'human computer' leading to the 'Turing machine'

Turing-computable (total) functions

Definition

A total function $f : \mathbb{N}^k \to \mathbb{N}$ is Turing-computable if there exists a Turing machine $M = \langle Q, \Sigma, \Gamma, \delta, q_0, \not{b}, F \rangle$ and a calculable coding function $\langle \cdot \rangle : \mathbb{N} \to \Sigma^*$ such that:

- for all $n_1, \ldots, n_k \in \mathbb{N}$ there exists $q \in F$ such that:

$$q_0 \langle n_1 \rangle \not{b} \langle n_2 \rangle \not{b} \ldots \not{b} \langle n_k \rangle \vdash_M^* q \langle f(n_1, \ldots, n_k) \rangle$$

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Recursive Functions

Functions defined by recursive equations:

like e.g. functions $+, \cdot, (\cdot)^{\cdot} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, and $(\cdot)! : \mathbb{N} \to \mathbb{N}$:

$$n + 0 = n$$

$$n + (m + 1) = (n + m) + 1$$

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$$n^0 = 1$$

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μ-Recursive (partial recursive) functions:

extend the primitive recursive functions by a μ-operator

that allows to obtain partial functions

Rósza Péter

Rósza Péter (1905–1977)

Primitive recursive functions ($\mathbb{N}^k \to \mathbb{N}$)

Base functions:

- $\mathcal{O} : \mathbb{N}^0 = \{\varnothing\} \to \mathbb{N}$, $\varnothing \mapsto 0$ (0-ary constant-0 function)
- $\mathsf{succ} : \mathbb{N} \to \mathbb{N}$, $x \mapsto x + 1$ (successor function)
- $\pi_i^n : \mathbb{N}^n \to \mathbb{N}$, $\vec{x} = \langle x_1, \ldots, x_n \rangle \mapsto x_i$ (projection function)

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Closed under operations:

- composition: if $f : \mathbb{N}^k \to \mathbb{N}$, and $g_i : \mathbb{N}^n \to \mathbb{N}$ are prim. rec., then so is $h = f \circ (g_1 \times \ldots \times g_k) : \mathbb{N}^n \to \mathbb{N}$:
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- primitive recursion: if $f : \mathbb{N}^n \to \mathbb{N}$, $g : \mathbb{N}^{n+2} \to \mathbb{N}$ are prim. rec., then so is $h = \mathsf{pr}(f; g) : \mathbb{N}^{n+1} \to \mathbb{N}$:
$$h(\vec{x}, 0) = f(\vec{x})$$
$$h(\vec{x}, y + 1) = g(\vec{x}, h(\vec{x}, y), y)$$

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Closed under operations:

- composition: if $f : \mathbb{N}^{km} \to \mathbb{N}^l$, and $g_i : \mathbb{N}^n \to \mathbb{N}^m$ are prim. rec., then so is $h = f \circ (g_1 \times \ldots \times g_k) : \mathbb{N}^n \to \mathbb{N}^l$:
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- $\pi_i^n : \mathbb{N}^n \to \mathbb{N}$, $\vec{x} = \langle x_1, \ldots, x_n \rangle \mapsto x_i$ (projection function)
- for $n > 1$: $\mathrm{id}^n : \mathbb{N}^n \to \mathbb{N}^n$, $\vec{x} = \langle x_1, \ldots, x_n \rangle \mapsto \vec{x}$ (n-ary identity f.)

Closed under operations:

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- primitive recursion: if $f : \mathbb{N}^n \to \mathbb{N}^l$, $g : \mathbb{N}^{n+l+1} \to \mathbb{N}^l$ are prim. rec., then so is $h = \mathsf{pr}(f; g) : \mathbb{N}^{n+1} \to \mathbb{N}^l$:
$$h(\vec{x}, 0) = f(\vec{x})$$
$$h(\vec{x}, y + 1) = g(\vec{x}, h(\vec{x}, y), y)$$

Primitive recursive functions (exercises)

Exercise

Show that the following functions are primitive recursive:

- addition
- constant functions
- multiplication
- (positive) sign-function
- the representing functions $\chi_=$ and $\chi_<$ for the predicates $=$ and $<$.

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Try-yourself-Examples

Show that the following functions are primitive recursive:

- exponentiation
- factorial

Admissible operations for primitive recursive functions

Proposition

1. *definition by case distinction:*

$$f(\vec{x}) := \begin{cases} f_1(\vec{x}) & \ldots P_1(\vec{x}) \\ f_2(\vec{x}) & \ldots P_2(\vec{x}) \wedge \neg P_1(\vec{x}) \\ \ldots & \\ f_k(\vec{x}) & \ldots P_k(\vec{x}) \wedge \neg P_{k-1}(\vec{x}) \wedge \ldots \wedge \neg P_1(\vec{x}) \\ f_{k+1}(\vec{x}) & \ldots \neg P_k(\vec{x}) \wedge \ldots \wedge \neg P_1(\vec{x}) \end{cases}$$

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2. *definition by bounded recursion:*

$$\mu z_{\leq y}.\, [P(x_1, \ldots, x_n, z)] := \begin{cases} z & \ldots\ \neg P(x_1, \ldots, x_n, i) \text{ for } 0 \leq i < z \leq y, \\ & \quad \text{and } P(x_1, \ldots, x_n, z) \\ y+1 & \ldots\ \neg \exists z. \big(0 \leq z \leq y \wedge P(x_1, \ldots, x_n, z) \big) \end{cases}$$

Properties of primitive recursive functions

Proposition

1. *Every primitive recursive function is total.*
2. *Every primitive recursive function is Turing-computable.*

Properties of primitive recursive functions

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1. *Every primitive recursive function is total.*
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Proof.

For (2):

- the base functions are Turing-computable
- the Turing-computible functions are closed under the schemes composition and primitive recursion

□

Turing-computable (total) functions

Definition

A total function $f : \mathbb{N}^k \to \mathbb{N}$ is Turing-computable if there exists a Turing machine $M = \langle Q, \Sigma, \Gamma, \delta, q_0, \not{b}, F \rangle$ and a calculable coding function $\langle \cdot \rangle : \mathbb{N} \to \Sigma^*$ such that:

- for all $n_1, \ldots, n_k \in \mathbb{N}$ there exists $q \in F$ such that:

$$q_0 \langle n_1 \rangle \not{b} \langle n_2 \rangle \not{b} \ldots \not{b} \langle n_k \rangle \vdash_M^* q \langle f(n_1, \ldots, n_k) \rangle$$

Features of computationally complete MoC's present?

- storage (unbounded)
- control (finite, given)
- modification
  - of (immediately accessible) stored data
  - of control state
- conditionals
- loop (unbounded)
- stopping condition

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Not primitive recursive (I)

Proposition

There exist calculable/Turing-computable functions that are not primitive recursive.

Proof.

By diagonalisation. □

Not primitive recursive (II): Ackermann function

Wilhelm Ackermann (1896–1962)

Not primitive recursive (II): Ackermann function

Ackermann function $\mathsf{A} : \mathbb{N}^2 \to \mathbb{N}$ (simplified version by Rósza Péter):

$$\mathsf{A}(0, x) = \mathsf{Succ}(x)$$

$$\mathsf{A}(x+1, 0) = \mathsf{A}(x, \mathsf{Succ}(0))$$

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A is not primitive recursive, it grows too fast:

$$\mathsf{A}(0, n) = n + 1$$

$$\mathsf{A}(1, n) = n + 2$$

$$\mathsf{A}(2, n) = 2n + 3$$

$$\mathsf{A}(3, n) = 2^{n+3} - 2$$

$$\mathsf{A}(4, n) = \underbrace{2^{2^{\cdot^{\cdot^{\cdot^{2^{16}}}}}}}_{n} - 3$$

$$\ldots$$

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For every primitive recursive $f : \mathbb{N} \to \mathbb{N}$ there exists some $i \in \mathbb{N}$ such that $f(i) < \mathsf{A}(i, i)$.

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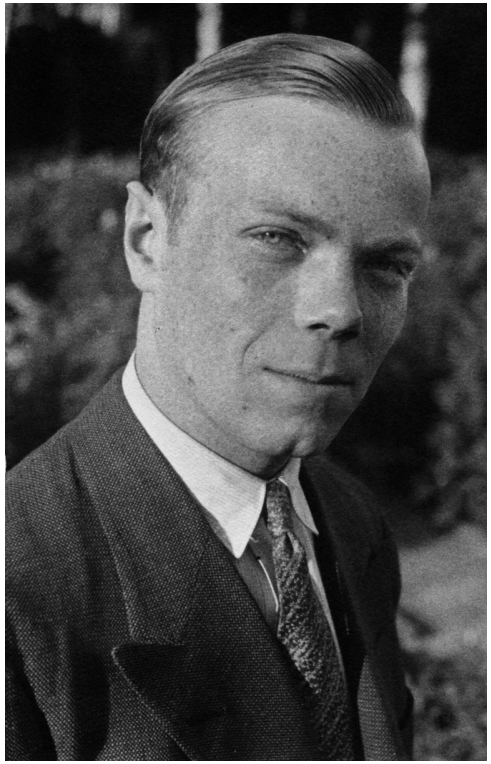
Theorem (Green, 1964)

For all $n \in \mathbb{N}$:

$$\mathsf{A}(n, n) \leq \mathit{BB}(2n)\,.$$

Jacques Herbrand



Jacques Herbrand (1908–1931)

Kurt Gödel

Kurt Gödel (1906–1978)

Gödel–Herbrand general recursive function

Defined by systems of recursion equations like that for the Ackermann function:

$$\mathsf{A}(0, y) = \mathsf{Succ}(y)$$

$$\mathsf{A}(\mathsf{Succ}(x), 0) = \mathsf{A}(x, \mathsf{Succ}(0))$$

$$\mathsf{A}(\mathsf{Succ}(x), \mathsf{Succ}(y)) = \mathsf{A}(x, \mathsf{A}(\mathsf{Succ}(x), y))$$

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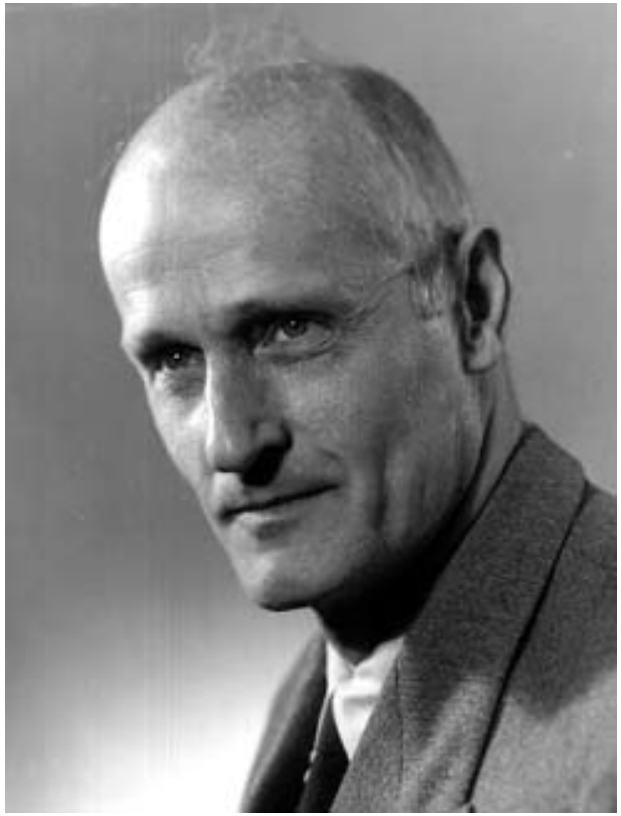
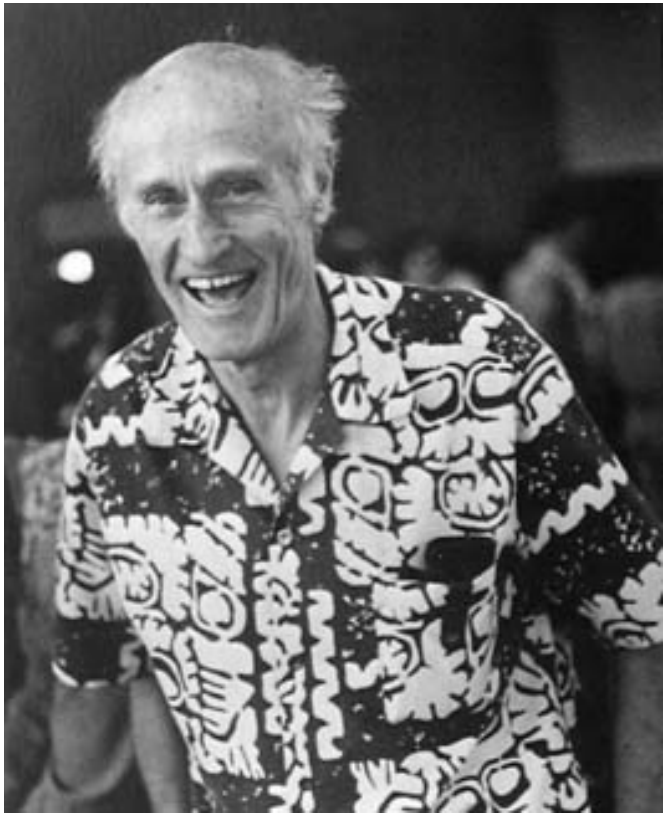
Numerals: $\langle 0 \rangle := 0$, and $\langle n \rangle := \underbrace{\mathsf{Succ}(\ldots \mathsf{Succ}(0))}_{n}$ for $n > 1$.

Definition

A function $f : \mathbb{N}^k \to \mathbb{N}$ is called general recursive if it can be defined by (such a) system $\mathcal{S}$ of recursion equations via a function symbol F if for all $n_1, \ldots, n_k \in \mathbb{N}$, the expression $F(\langle n_1 \rangle, \ldots, \langle n_k \rangle)$ evaluates according to $\mathcal{S}$ to a unique numeral $\langle n \rangle$, and such that furthermore: $n = f(n_1, \ldots, n_k)$.

Stephen Cole Kleene





Stephen Cole Kleene (1906–1994)

Unbounded minimisation (μ-recursion)

Let $f : \mathbb{N}^{k+1} \to \mathbb{N}$ total. Then the partial function defined by:

$$\mu(f) : \mathbb{N}^k \rightharpoonup \mathbb{N}$$

$$\vec{x} \mapsto \begin{cases} \min\{y \in \mathbb{N} \mid f(\vec{x}, y) = 0\} & \ldots \exists y\,(f(\vec{x}, y) = 0) \\ \uparrow & \ldots \text{else} \end{cases}$$

is called the unbounded minimisation of f.

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Let $f : \mathbb{N}^{k+1} \rightharpoonup \mathbb{N}$ partial. Then the partial function $\mu(f)$:

$$\mu(f) : \mathbb{N}^k \rightharpoonup \mathbb{N}$$

$$\vec{x} \mapsto \begin{cases} z & \ldots f(\vec{x}, z) = 0 \;\wedge\; \forall y\,\big(0 \leq y < z \to (f(\vec{x}, y)\downarrow \neq 0)\big) \\ \uparrow & \ldots \neg\exists y\,\big(f(\vec{x}, y) = 0 \;\wedge\; \forall z\,(0 \leq z < y \to (f(\vec{x}, z)\downarrow)\big) \end{cases}$$

is called the unbounded minimisation of f.

Partial, and total, recursive functions

Definition

A partial function $f : \mathbb{N}^n \rightharpoonup \mathbb{N}^l$ is called partial recursive if it can be specified from base functions ($\mathcal{O}$, succ, π_i^n, and id^n) by successive applications of composition, primitive recursion, and unbounded minimisation.

A partial recursive function is called (total) recursive if it is total.

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A partial recursive function is called (total) recursive if it is total.

Proposition

Every partial recursive function is Turing-computable.

Primitive recursive

- storage (unbounded) ✓
- control (finite, given) ✓
- modification ✓
  - of (immediately accessible) stored data
  - of control state
- conditionals ✓
- loop ✓ (unbounded) ✗
- stopping condition ✓

Partial recursive = prim. rec. + unbounded minimization

- storage (unbounded) ✓
- control (finite, given) ✓
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Turing-computable functions

Definition

1. A total function $f : \mathbb{N}^k \to \mathbb{N}$ is Turing-computable if there exists a Turing machine $M = \langle Q, \Sigma, \Gamma, \delta, q_0, \not{b}, F \rangle$ and a calculable coding function $\langle \cdot \rangle : \mathbb{N} \to \Sigma^*$ such that:
   - for all $n_1, \ldots, n_k \in \mathbb{N}$ there exists $q \in F$ such that:

$$q_0 \langle n_1 \rangle \not{b} \langle n_2 \rangle \not{b} \ldots \not{b} \langle n_k \rangle \vdash_M^* q \langle f(n_1, \ldots, n_k) \rangle$$

Turing-computable functions

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Partial recursive vs. Turing-computable functions

Lemma

Every Turing-computable function is partial recursive.

Partial recursive vs. Turing-computable functions

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Every Turing-computable function is partial recursive.

Proof by arithmetization of Turing machines, showing:

Theorem (Kleene's normal form theorem)

For every Turing-computable, partial function (and hence for every partial recursive function) $h : \mathbb{N}^k \to \mathbb{N}$ there exist primitive recursive functions $f : \mathbb{N} \to \mathbb{N}$ and $g : \mathbb{N}^{k+1} \to \mathbb{N}$ such that:

$$h(x_1, \ldots, x_n) = (f \circ \mu(g))(x_1, \ldots, x_n)$$

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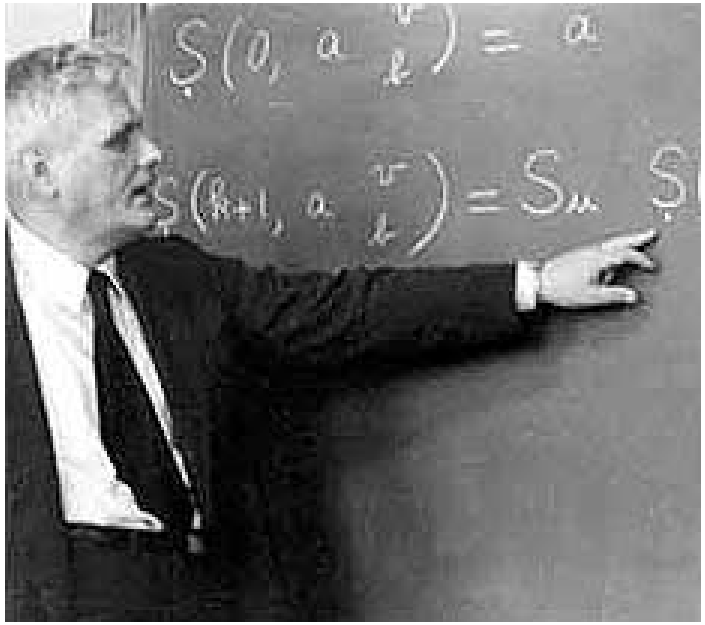
Theorem

The Turing-computable (partial) functions coincide with the partial recursive functions.

Alonzo Church



Alonzo Church (1903–1995)

Effectively calculable functions

Alonzo Church (1936):

> *"We now define the notion [. . .] of an effectively calculable function of positive integers by identifying it with the notion of a recursive function of positive integers (or a λ-definable function of positive integers). This definition is thought to be justified by the considerations which follow, so far as positive justification can ever be obtained for the selection of formal definition to correspond to an intuitive notion."*

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Definition (Church)

For every total function $f : \mathbb{N} \to \mathbb{N}$, and partial function $g : \mathbb{N} \rightharpoonup \mathbb{N}$,

$$f \text{ is effectively calculable } :\iff f \text{ is recursive}$$

$$g \text{ is effectively calculable } :\iff g \text{ is partial-recursive}$$

Church's Thesis

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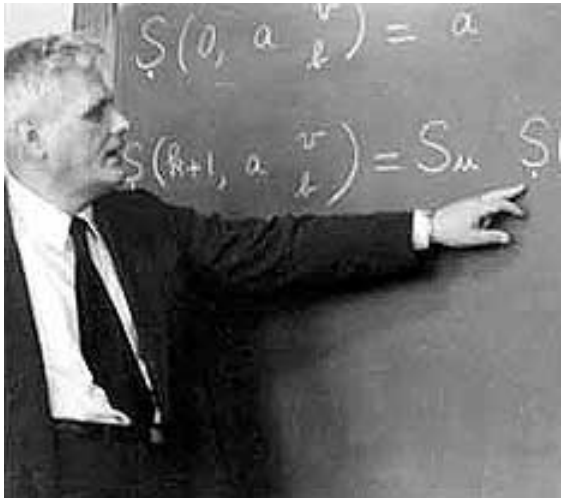
λ-calculus



Alonzo Church (1903 –1992)

Theorem (Kleene/Church, 1935)

Every λ-definable function is general recursive, and vice versa.

Recommended reading

1. Recursive and primitive-recursive functions:

   Chapter 3, The Lambda Calculus of the book:

   - Maribel Fernández [4]: *Models of Computation (An Introduction to Computability Theory)*, Springer-Verlag London, 2009.

Post's 'working hypothesis'

E.L. Post in his 1936 article (Post machines):

> *"The writer expects the present formulation to turn out to be logically equivalent to recursiveness in the sense of the Gödel–Church development. Its purpose, however, is not only to present a system of a certain logical potency but also, in its restricted field, of psychological fidelity. In the latter sense wider and wider formulations are contemplated. On the other hand, our aim will be to show that all such are logically reducible to formulation 1 [Post machines]. We offer this conclusion at the present moment as a working hypothesis. And to our mind such is Church's identification of effective calculability with recursiveness."*

Church on Post's 'working hypothesis'

Alonzo Church in his review (1937) of Post's 1936 article:

> *"The author proposes a definition of "finite 1-process" which is similar in formulation, and in fact equivalent, to computation by a Turing machine (see the preceding review). He does not, however, regard his formulation as certainly to be identified with effectiveness in the ordinary sense, but takes this identification as a "working hypothesis" in need of continual verification. To this the reviewer would object that effectiveness in the ordinary sense has not been given an exact definition, and hence the working hypothesis in question has not an exact meaning. To define effectiveness as computability by an arbitrary machine, subject to restrictions of finiteness, would seem to be an adequate representation of the ordinary notion, and if this is done the need for a working hypothesis disappears."*

Church on Turing's paper

A. Church in his review (1937) of Turing's 1936 article:

> *"The author proposes as a criterion that an infinite sequence of digits 0 and 1 be "computable" that it shall be possible to devise a computing machine, occupying a finite space and with working parts of finite size, which will write down the sequence to any desired number of terms if allowed to run for a sufficiently long time. As a matter of convenience, certain further restrictions are imposed on the character of the machine, but these are of such a nature as obviously to cause no loss of generality—in particular, a human calculator, provided with pencil and paper and explicit instructions, can be regarded as a kind of Turing machine. It is thus immediately clear that computability, so defined, can be identified with (especially, is no less general than) the notion of effectiveness as it appears in certain mathematical problems [. . .].*

Busy Beaver and axiomatic-theory consistency

Proposition

Let T be a computable and arithmetically sound axiomatic theory. Then there exists a constant n_T such that for all $n \geq n_T$, no statement of the form "BB(n) = k" can be proved in T.

Busy Beaver and axiomatic-theory consistency

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There is an explicit 748-state Turing machine that halts iff the set theory ZF is inconsistent. Hence, assuming that ZF is consistent, ZF cannot prove the value of BB(748).

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Theorem (GitHub user "Code Golf Addict")

There is an explizit 27-state Turing machine that halts iff Goldbach's Conjecture is false.

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Theorem (Matiyasevich, O'Rear, Aaronson)

There is an explicit 744-state Turing machine that halts iff the Riemann Hypothesis is false.

Summary

Recursive functions

- primitive recursive functions
- Gödel–Herbrand(–Kleene) general recursive functions
- partial recursive functions
  - defined with μ-recursion (unbounded minimisation)
- Partial recursive functions = Turing computable functions
- Church's thesis
  - effectively calculable functions $\hat{=}$ partial-recursive functions
  - some debate

Course overview

intro	*classic models*			*additional models*
Introduction to Computability	**Machine Models**	**Recursive Functions**	**Lambda Calculus**	**Three more Models of Computation**
computation and decision problems, from logic to computability, overview of models of computation relevance of MoCs	Post Machines, typical features, Turing's analysis of human computers, Turing machines, basic recursion theory	primitive recursive functions, Gödel–Herbrand recursive functions, partial recursive funct's, partial recursive = = Turing-computable, Church's Thesis	λ-terms, β-reduction, λ-definable functions, partial recursive = λ-definable = Turing computable	Post's Correspondence Problem, Interaction-Nets, Fractran
	imperative programming	*algebraic programming*	*functional programming*	

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on youtube at `https://youtu.be/OST1DjD08Hg?si=PS9O0x3szKF7LhSZ`.

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